Lung-Focused Inputs for CNN-ViT Pneumonia Classification

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Computer Vision Project / Final Milestones

Abstract

We reconstruct a five-CNN pneumonia classifier and evaluate an anatomy-guided extension using a resource-bounded frozen-feature pilot. The historical five-epoch ensemble attained 92.15% accuracy and ROC AUC 0.9623 on 624 benchmark records. The final experiment compares full and lung-focused images with CNN-only and CNN-plus-transformer classifiers using validation-selected settings. We report all measured conditions, including unfavorable outcomes, rather than assuming anatomical cropping improves classification.

In the final matched pilot, full-image CNN accuracy was 86.86%, lung-crop CNN accuracy was 92.63%, and the crop CNN/ViT candidate accuracy was 93.75%. The measured crop macro-F1 change was +0.0696.

I. Introduction and Related Work

Pediatric pneumonia recognition from radiographs is a binary benchmark task with important limitations in generalization. Chouhan et al. combined five ImageNet CNNs [1]. TorchXRayVision provides pretrained anatomical segmentation [2]. DeiT supplies pretrained data-efficient image transformers [3]. Anatomical CNN–Transformer hybrids already exist [4]. Our contribution is a controlled student reproduction and pilot evaluation, not a new clinical method.

II. Dataset and Historical Reconstruction

We use the public pediatric chest X-ray dataset [5]. The prior reconstruction trained five CNNs on 5,232 records for five epochs. Majority voting produced 206 true negatives, 28 false positives, 21 false negatives and 369 true positives on 624 records. The corresponding sensitivity was 94.62%, specificity 88.03%, precision 92.95% and macro F1 0.9157. This schedule is shorter than the selected paper. Fresh augmentation was already part of this baseline.

III. Final Experiment

The primary hypothesis proposed macro-F1 improvement of 0.01–0.03 and specificity improvement of 2–5 percentage points from anatomical cropping. A separate transformer extension tests complementary predictions. Segmentation uses 256-pixel input rather than the standard 512-pixel wrapper to bound CPU runtime. Its mask defines a crop on original source pixels. The pilot uses 4,164 training and 1,042 validation records, with 624 original test records retained. It uses frozen pretrained encoders and five-epoch classifier-head optimization, which differs from online late-block training in M2. Full and cropped conditions share a split and comparable head training. Validation data select checkpoints and mixtures. Benchmark test predictions do not select settings.

All head experiments use CPU FP32 with two threads and seed 42. AdamW optimizes unweighted cross-entropy for five epochs at learning rate 0.001, weight decay 0.01 and batch size 32. Feature standardization uses training-only means and standard deviations, with a 0.001 standard-deviation floor. Encoder inputs use ImageNet channel normalization. Checkpoint ties favor the earlier epoch; no fresh image augmentation is used in this declared frozen-feature pilot.

IV. Results

A: accuracy 86.86%, recall 97.44%, specificity 69.23%, macro F1 0.850, AUC 0.931

B: accuracy 92.63%, recall 96.67%, specificity 85.90%, macro F1 0.920, AUC 0.970

C: accuracy 88.14%, recall 97.95%, specificity 71.79%, macro F1 0.866, AUC 0.947

D: accuracy 93.75%, recall 98.21%, specificity 86.32%, macro F1 0.932, AUC 0.976

On the 618 unique test images, D achieved 93.69% accuracy, 0.9310 macro F1 and 0.9755 AUC. The selected ViT probability weight was 0.25 for C and 0.50 for D; the remaining weight belongs to the corresponding CNN. Labels use a fixed 0.5 threshold.

The measured macro-F1 gain exceeds the predicted +0.03 upper bound. Crop macro-F1 difference B minus A: 0.06959. Paired interval: [0.0404, 0.1016]. Specificity change: 16.667 percentage points.

The measured interval between the first and last uniform 256 px mask files was 102.7 minutes. This includes inter-image pauses and later shard downloads but excludes the initial model and first-shard downloads. Head-training time was not independently captured.

Classifier parameter counts (excluding the fixed segmenter): resnet18_encoder: 11,176,512, resnet18_full_head: 1,026, resnet18_crop_head: 1,026, deit_tiny_encoder: 5,524,416, deit_tiny_full_head: 386, deit_tiny_crop_head: 386.

Validation macro F1 was A 0.9549, B 0.9072, C 0.9573, D 0.9243. Cropping reduced validation macro F1 despite improving benchmark test macro F1. The direction reverses across these partitions. This single-split experiment therefore does not establish robust crop superiority; the cause of this reversal is unresolved. D was the predeclared target condition, not a winner selected across conditions using test scores.

V. Failure Analysis and Limitations

Automatic geometry and empty-mask checks used the full-image fallback for 82 of 5824 unique images (1.41%). The median retained crop area was 71.1% of the original area. This is an automatic plausibility audit, not a measured segmentation error rate: manual mask annotations are unavailable.

False positives and false negatives should be assessed together. Anatomical crops may exclude relevant tissue, and pretrained segmentation may transfer imperfectly to pediatric images. Filename groups are imperfect patient identifiers. Duplicate-image sensitivity analysis and paired intervals improve transparency but do not establish external generalization. The benchmark test was inspected during prior work and is not a blind external holdout. Uncalibrated probability scores are not clinical risk estimates. Grad-CAM provides qualitative component attribution without lesion ground truth. The reduced pilot does not establish performance of the original full five-CNN-plus-ViT proposal.

The fixed anatomical segmenter contains 68,068,316 parameters. Including the two encoders and two D heads, the complete D pipeline contains 84,770,656 neural-network parameters; only 1,412 classifier-head parameters are optimized for D. The segmenter is therefore a substantial part of total compute, rather than a free preprocessing operation.

Measured server-side processing for two cached prepared samples was 1.49 and 1.24 seconds. One uploaded 224-pixel fixture took 4.58 seconds, including 3.52 seconds of preprocessing and 1.03 seconds of classifier processing. Timings include model loads and image encoding where applicable and exclude HTTP transport. These are fixture measurements, not latency guarantees.

VI. Conclusion

The predeclared crop CNN/ViT condition achieved 93.75% accuracy, 98.21% sensitivity and AUC 0.9757 on 624 benchmark records. The primary crop macro-F1 gain exceeded the predicted range, while the validation reversal limits generalization. The project delivers a runnable model and a reproducible resource-bounded comparison; repeated patient-aware splits and independent external data remain necessary.

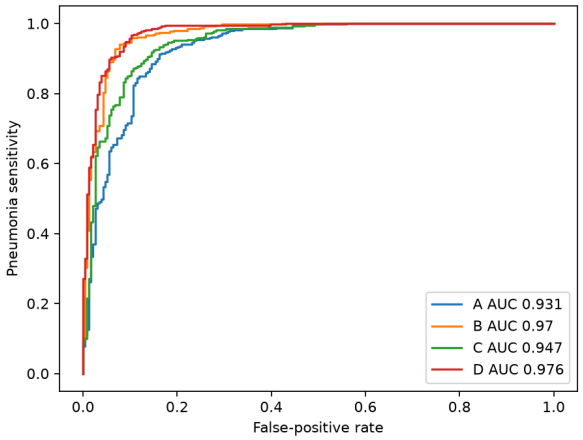

Figure 1. ROC curves for the four declared conditions on 624 test records.

References

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